



BLOCKCHAIN LAYER 2 ROLLUPS, OPTIMISTIC VS ZERO KNOWLEDGE

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ABSTRACT

The current Layer 1 blockchain, such as Ethereum and Bitcoin, has several problems related to scalability and cost. Layer 1 blockchain networks have always been a struggle for Ethereum over scalability. Many of these factors can be attributed to the network's nature of linear processing of transactions. During high traffic, these networks get crowded, slow the transaction time, and charge high fees. For many industries, these have made blockchains unusable, even though previously, I described them as getting from bad to worse in terms of functionality. To tackle these issues, developers have a singular aim in mind – to craft blockchain scaling solutions that improve efficiency, expand capacity, and bring down costs of transactions so that blockchains can deliver the high performance demanded of them to match today's digital world.

Rollups have become one of the most promising innovations in satisfying the stated need. Rollups can be described as a particular class of blockchains whose primary function is to increase the transactional capacity. Rollups, in the case of blockchain, are 'layer 2' scaling solutions that improve the transaction rate of the blockchain network. Thus, "Rollups" accurately name what they do – they combine several transactions into batches outside the mainstream network before performing these operations as one transaction in the mainnet.

Keywords: Blockchain Scalability, Layer 1 Blockchain, Transaction Fees, Rollups, Layer 2 Solutions

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INTRODUCTION

Business blockchain has rapidly developed over the past years because it enhances decentralization, transparency, and security in organizations. This has exposed important scaling challenges, especially in the most used networks, such as Ethereum and Bitcoin. Due to the linear flow of transactions in such networks, there is always a bottleneck in periods of high traffic, leading to high transaction rates and fees. These problems act as hurdles and significantly inhibit large-scale blockchain technology adoption.

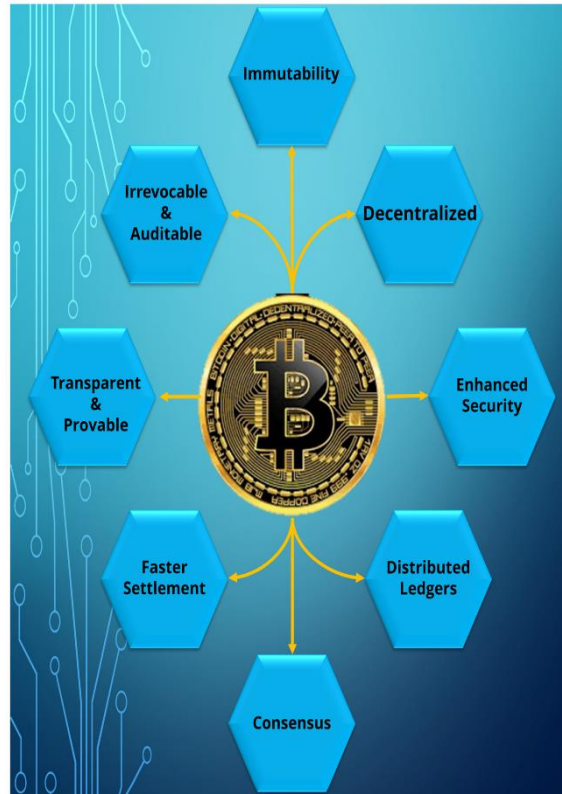


Figure 1 Blockchain system

Developers have suggested several scaling solutions to address these difficulties. Of these, Layer 2 solutions are promising, especially rollups. Rollups are designed to settle transactions off the main chain and summarize several micro trades into one macro trade. This increases transaction capability throughput and consequently results in cost reduction.

Base layers, or the first layer of blockchains, are designed to be highly secure and completely decentralized. However, these base layers have some scalability issues because of the consensus algorithms like POW or POS. However, these blockchains are justified in slowing down as more users and transaction requests are provided, and their fees rise. He noted that it could be better for decentralized applications (dApps), which need cheap and fast transactions for a good user experience.

These problems are solved in Rollups by transferring most of the work related to the transaction to a separate chain. Thus, they can have better throughputs and costs at the same time without compromising on the security that comes from the blockchain. Composed transactions collect a number of transactions into one batch, and this batch is credited to the main chain. This cuts down on the information that has to be placed in the main chain, hence enhancing scalability.

There are two main types of rollups: Unwinding Rollups, idealistic Rollups, and Zero-Knowledge (ZK) Rollups. They differ in how they handle and check transaction details, and they each have advantages and, of course, some disadvantages.

Optimistic Rollups work on the principle that all transactions are acceptable and resolve the transactions off the main chain (Adler & Quintyne-Collins, 2019). These are posted to the main chain without initial proof of validity, so they are not effectively protected from double-spending. This negates the computational problem and makes the transactions fast and cheap. However, it involves using a stewards' model of validation and provoking a dispute time if the validators think there is an error or fraud.

Whereas in Zero-Knowledge Rollups, completeness and correctness of the transaction are guaranteed by cryptographic proofs. This means that every set of transactions taking place off the main chain will come with evidence and will need to be checked on the main chain. This method offers the immediate final and highly protected commitment with the disadvantage of cryptographic calculations, which may entail high costs.

Each kind of rollup significantly enhances scalability, throughput, and cost compared to the native Layer 1 blockchain transactions. By handling the limitations of base-layer blockchains, rollups make it possible to include blockchain technology in many causes and integrated uses, including decentralized finance (DeFi), online gaming, logistics and supply, and identity services.

Rollups are predicted to contribute to the improvements of blockchain networks in the future developments of the blockchain ecosystem (Montanari, 2020). More work needs to be done to fine-tune rollup solutions and solve the remaining issues concerning security, compatibility, and ease of use. In this way, through optimizing the rollups' benefits, blockchain networks will have the opportunities and prospects to realize the increase in scalability and efficiency for developing the subsequent generation of apps and services.

Types of Rollups:

Rollups can be broadly classified into two main types: There are two types of Rollups, namely Optimistic Rollups and Zero-Knowledge (ZK) Rollups (Steinkopf, 2020). Both types have their way of handling and authenticating the transactions, and they also have their qualities and cons.



Figure 2 Optimistic Rollups Vs Zero-Knowledge Rollups

Optimistic Rollups:

It should also be noted that both Optimistic Rollups work on the premise that all transactions are highly legitimate. These are executed and settled off-chain without first proving their validity on the main chain (Steinkopf, 2020). This approach helps considerably offload computational work from the main chain, thus making the transactions faster and cheaper.

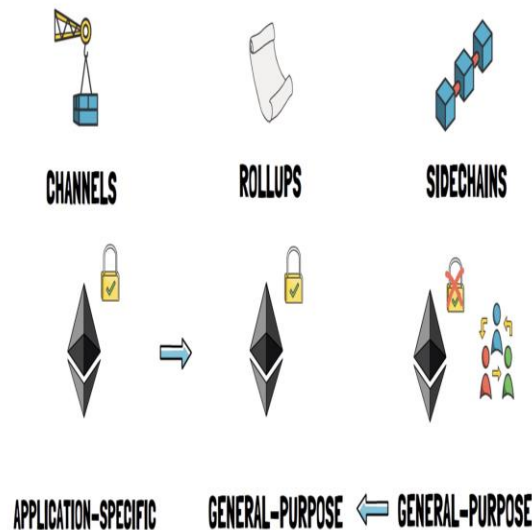


Figure 3 rollups

ADVANTAGES OF OPTIMISTIC ROLLUPS:

Efficiency:

Optimistic Rollups provide vast efficiency benefits since they lessen the work performed on the main chain. Most of the actual transaction workflows are conducted off-chain, which makes it beneficial for managing the critical network's efficiency and steadiness. This proves useful as it relieves the main chain of bearing the load of these services, allowing it to focus on other essential roles such as consensus and security (Swanson, 2015). Also, transaction processing is optimally distributed within the network through Optimistic Rollups. The result may be improved on-chain throughput and, hence, reduced transaction latency and confirmation times.

Scalability:

Concerning scalability, Optimistic Rollups make it possible to engage more transactions per second because the primary handling of transactions does not occur on the chain. This makes it possible for the network to facilitate a very high TPS compared to what can be offered by the traditional on-chain systems. It is faster by way of improved throughput, and users will be able to confirm numbers of transactions much quicker, thus improving the UI/UX of highly interaction-based applications, such as decentralized finance (DeFi) and gaming applications. Both optimistic rolls have the potential to handle thousands of TPS depending on the general architecture and the network used. To borrow Hand's analogy of scalability, it is crucial to meet the increased demand for blockchain technologies. In the ever-evolving blockchain space scenarios, the optimization capability of Optimistic Rollups ensures that it is adaptable to take larger groups and extensive amounts of transactions on a network without hurting functionality.

Cost:

Yet another advantage of Optimistic Rollups is the cost elements that are greatly optimized. This means that the number of transactions that can be conducted within the network would significantly increase. At the same time, the transaction fees would be lowered due to the decrease in resource usage since most of the computational work is offloaded to the nodes, consequently cutting down the overall costs of the system. This makes Optimistic Rollups an efficient tool for developers and an affordable one for users. Economies of scale work hand in hand with cost levels in that as the number of transactions increases, the cost per transaction, in turn, reduces, making it possible for the user to engage more in the blockchain. Lower gas fees mean users and consumers can pay a manageable cost for making transactions and using decentralized applications (dApps) on the blockchain; hence, they are used more often. Thus, by cutting the cost of transaction operations, Optimistic Rollups will attract attention to the increased use of blockchain networks and various services and applications based on them (Gabizon et al., 2020).

Development and Deployment:

Optimistic rollups even reduce complexity in the development and deployment processes. Due to this, their challenge-response mechanism is relatively easy to deploy and integrate on an existing blockchain, creating a more accessible environment for developing scalable solutions. Because of the layperson interpretation in Optimistic Rollups, there are quicker ways in which scalable solutions can be launched to the market, giving projects the ability to take their applications to the next level. Due to the modularity of Optimistic Rollups, various solutions can be employed according to the exclusive characteristics of the rollup architecture. The roll-up architecture can also be made incrementally with upgrades and improvements done with evolution without fundamental changes to the blockchain network (Chen et al., 2020).

Security:

Security is a significant strength of Optimistic Rollups; there is a challenging period where any participant can challenge a transaction (Mattessich & Johnson, 2018). This stage protects against scams or any invalid transaction attempts. It ties the validators' incentives directly to keeping an eye on the network and rejecting any transaction that is anything but valid, thus increasing the overall security and reliability of the rollup. Optimistic Rollups have safeguards to deal with any possible attacks and quality of transactions within the unit. In the continuously developing ecosystem, Optimistic Rollups can apply new measures of security to address the constantly rising threats.

Disadvantages:

Still, Optimistic Rollups have drawbacks, mainly associated with security and the speed of withdrawals.

Security:

A critical risk involving Optimistic Rollups is the need for a challenge period. It takes a few hours to a week, after which any of the parties can challenge a transaction. This mechanism is very crucial for the integrity of the Rollup since it would allow fraudulent or invalid transactions to be pointed out and rectified. The downside is that this brings in a wait before closure, so while a commodity can be purchased and sold, they are only considered a definite deal once the challenge is over. This is one of the significant disadvantages of using this technology since it may prove disadvantageous for any solution that needs fast settlement of transactions.

Similar to other Layer 2 solutions, the model of security of Optimistic Rollups is based on incentives (Al Bassam, 2020).. However, the stakeholders in the network are expected to observe the Rollup and report any transactions that ought to be queried by the Rollup. This is rather useful when it comes to preventing fraud, but at the same time, it shifts much of the responsibility onto the participation of the community. In this case, if the incentives are not strong enough, or there are not enough participants to perform the monitoring, then the security of Rollup can be threatened.

Withdrawal Time:

The other main disadvantage of Optimistic Rollups is that the withdrawal time it takes tends to be much longer because of the dispute period (Doty et al., 2015). Those who want to return their funds to the main chain must wait for the challenge times to guarantee that the corresponding transactions are non-refundable. Depending on the chosen rollup implementation and the network's availability, its duration may vary from several hours to a week. These delays may be inconvenient, especially for users who require their funds on the double; this may reduce the likelihood of using the Rollup in some cases that require high liquidity.

It makes the processes longer, and when the elimination takes longer, other risks and conditions may be augmented. When the hold is in place, users might experience fluctuations in the market that affect the value of their property. Moreover, the longer withdrawal time can become an issue for users who have to wait longer for finality than simple on-chain transactions, despite Optimistic Rollups being much better ranked.

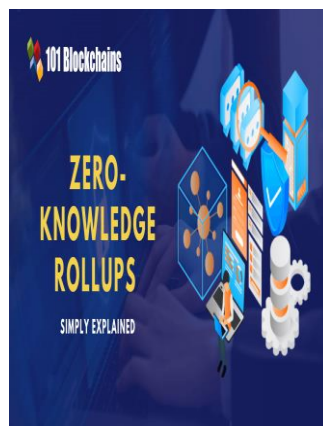
Optimistic Rollups have a variety of advantages regarding scalability, optimization, and reduction of costs (Dulloor et al., 2016). Yet, the drawbacks concern such aspects as security and the time taken to withdraw. The introduction of a challenge period slows the process before transaction completion. Besides, security needs to be underpinned by economic incentives, which may be flawed for applications that require immediate and reliable confirmation. Moreover, longer withdrawal times can be disadvantageous in such a case; the user wants his tokens back immediately, thus discouraging the usage of Optimistic Rollups for such applications.

Table 1 Key Features and Use Cases of Optimistic Rollups

Category	Details
Fraud Proofs	- Optimistic Rollups use fraud proofs to ensure transaction integrity. - If disputed, a fraud proof is submitted to the main chain to prove invalidity. - Validators/watchers are incentivized to monitor and challenge invalid transactions, maintaining rollup security.
Economic Incentives	- Participants are economically incentivized to act honestly. - Malicious actors risk losing their stake/bond if they attempt invalid transactions. - Validators who successfully dispute fraudulent transactions are rewarded, encouraging vigilant monitoring.
Use Cases	- DeFi applications benefit from higher throughput and lower fees. - Gaming and NFT platforms enable faster, more affordable transactions for users.
Examples	- Arbitrum and Optimism are prominent implementations. - Both projects aim to enhance Ethereum's scalability while maintaining security and decentralization.
Interoperability	- Designed to be compatible with existing Ethereum smart contracts. - Allows developers to deploy applications on rollups with minimal modifications. - Ensures benefits can be leveraged without significant codebase changes.

Zero-Knowledge (ZK) Rollups:

Instantiations of ZK Rollups include Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (zk-SNARKs) and Zero-Knowledge Scalable Transparent Arguments of Knowledge (zk-STARKs) with outputs that comprise the validity proofs of the transactions. Every set of transactions conducted in batches of the chain is associated with a proof placed on the chain. The main chain then checks this proof to confirm the validity of all the other transactions contained in the batch (Lewenberg et al., 2015).

**Figure 4** Zero-Knowledge Rollups

ADVANTAGES:

Security:

Another combined perk is the fact that ZK Rollups are secure. ZK Rollups also give Finality with cryptographic assurances to the security of transactions, so they are immutable. Zero-knowledge proofs are completely safe and provide high levels of anonymity; non-disclosed details of single operations are obscured. Due to these reasons, ZK Rollups are well suited for applications that heighten the degree of privacy and security.

Efficiency:

Regarding efficiency, the same as in the case of transaction membership proof, ZK Rollups allows for fast transaction finality; there is no need for a challenge period because the validity proof is checked on the main chain. This does away with the challenge period that other rollup types, such as Optimistic Rollups, go through. Since the transactions are processed and validated by the rollup, ZK Rollups help to lessen the burden on the main chain computations. This allows the main chain to maintain efficiency and work on other objectives like agreement and security.

Scalability:

ZK Rollups are priced for high scalability in the transaction throughput level while requiring low storage data. This is because only the validity proof and the minimal transaction data are written to the main chain, much less than the total virtual chain space. Therefore, ZK Rollups would accommodate thousands of transactions in a single batch, thus improving the efficiency of blockchain networks' scalability. This type of scalability is essential for developing the blockchain product used for growing infrastructure demands for specific applications with corresponding services.

DISADVANTAGES

Complexity:

It was noted that ZK Rollups also has certain disadvantages, mainly connected with the issues of increased intricacy and costs. The general steps to implement ZK Rollups take longer and require much more work than others because of the various cryptographic methods used in this rollup type. Building and scaling ZK Rollup testaments requires technical skills in SSL, zero-knowledge proofs, and endorsers of cryptographic systems. This can be quite a problem, especially for developers and organizations who want to implement ZK Rollups successfully.

Cost:

The cost of deploying ZK Rollups can be slightly higher at the onset than in other types of rollups since some of them need to be conducted in trusted sets like in the zk-SNARKs. Trusted setups need a well-protected and efficient mechanism to produce cryptographic parameters; this adds more weight and volume to the solutions and, hence, is costly to implement. This process must be performed in a way to keep cryptographic parameters secure. There are no second chances for the trusted setup; hence, any security breach in this phase will compromise a more extensive system. Also, it was mentioned that zk-STARKs are agnostic to the trusted setup. Nevertheless, they are computation-demanding and require relevant skills to be performed.

Privacy and Proof Generation:

ZK Rollups can provide better privacy because transaction details can be hidden with the help of zero-knowledge proofs, but at the same time, it is possible to check their correctness. This makes ZK Rollups especially suitable for use cases that demand high privacy protection. Nonetheless, the generation of such proofs is computationally expensive, sometimes posing a challenge in terms of the resources to be used. The generated proofs are extremely efficient to verify once created and can thus be processed on the main chain.

Use Cases and Examples:

By using ZK Rollups, applications that require privacy and security of operations will be better placed as financial applications require confidential operations in transferring financial assets. SCM solutions and ID tracking can integrate ZK Rollups to obtain private and provable credentials or identities. Some properly established Integrated ZK Rollup projects include zkSync and StarkWare. While zkSync primarily uses zk-SNARKs, StarkWare comprises zk-STARKs in their rollup solutions. The two projects aim to improve the scalability, security, and privacy of Ethereum and other blockchains.

Interoperability and Economic Incentives:

ZK Rollups should be able to support old Ethereum intelligent contracts, so developers will not need to make many alterations to shift their projects to rollups. They pinpoint that this compatibility guarantees that the advantages of ZK Rollups can be utilized without substantial modifications to the codebases. In contrast with Optimistic Rollups, ZK Rollups do not depend on the bounties or the challenge period in making the transactions valid. It makes the security model less complex and only depends on other people, such as validators or watchers.

In conclusion, ZK Rollups can be one of the most effective tools to address blockchain systems' security, privacy, and scalability problems. Nonetheless, based on their enhancement and the superior security attributes that conform to intricate cryptographic algorithms and fast transaction processing, intelligent contracts present clear advantages that warrant their implementation in applications that may entail secure, immutable transactions.

Thus, due to the tendency to enhance the performance and availability of cryptographic techniques related to ZK Rollups, these technologies will remain relevant as the basis for future blockchain.

Table 2 Overview of ZK Rollups: Key Features and Use Cases

Aspect	Details
Privacy	- Enhanced privacy features by concealing transaction details while proving validity.
	- Particularly appealing for applications requiring strong privacy guarantees.
Proof Generation and Verification	- Computationally intensive proof generation requiring significant resources.
	- Proofs are quick and easy to verify, enabling efficient validation on the main chain.
Use Cases	- Financial applications benefit from confidential transactions and asset transfers.
	- Supply chain and identity verification systems use ZK Rollups for secure and private proofs of authenticity and identity.
Examples	- zkSync (uses zk-SNARKs) and StarkWare (uses zk-STARKs) are prominent ZK Rollup implementations.
	- Both aim to enhance scalability, security, and privacy of Ethereum and other blockchain networks.
Interoperability	- Designed to be compatible with existing Ethereum smart contracts, allowing minimal modifications for deployment.
	- Ensures benefits of ZK Rollups can be leveraged without significant changes to existing codebases.
No Need for Economic Incentives	- Unlike Optimistic Rollups, does not rely on economic incentives or challenge periods to ensure validity of transactions.
	- Simplifies the security model and reduces reliance on external validators or watchers.
Overall Benefits	- Enhanced security, privacy, and scalability for blockchain networks.
	- Well-suited for a wide range of applications requiring robust cryptographic guarantees.

Performance Analysis:

In a cohort with Layer 1 blockchain transactions, Optimistic and ZK Rollups help substantially improve the situation. However, they differ depending on their characteristics, which include transaction rate, costs, and security features.

Transaction Throughput:

Optimistic Rollups can process a throughput rate of thousands of TPS based on the type of implementation and the network's preparedness (Sherman, 2014). Their throughput is also tilted higher by decoupling transaction processing from the main chain, thus increasing scalability. However, one restriction that might affect the overall maximum possible TPS is the on-chain data availability constraints. Likewise, ZK Rollups indicate a similar capability to that of Optimistic Rollups. At the same time, some of the implementations may even be capable of providing the said throughput because of less complex proof validations. Transaction processing and zero-knowledge proofs are highly scalable due to the batch processing of transactions and the compact size of proofs. The main limitation of ZK Rollups is the computational cost of proof creation. However, cryptographic technologies are constantly getting optimized to be less compute-intensive.

Transaction Costs:

Measurement based on the transaction cost shows that Optimistic Rollups entail lower transaction fees because computations on the main chain are fewer.

Thus, users can save on gas costs that are considerably lower compared to Layer 1 transactions, although fees can depend on the transaction loads and the concrete type of rollup (Gudgeon et al., 2019). Conversely, the payments are slightly higher in ZK Rollups because of the computational proof that needs to be made cryptographically. Although proof generation is computationally more expensive than on-chain transactions, the transaction fees of ZK Rollups are much lower than those of everyday on-chain transactions. While defining the cost profile of the ZK Rollups, the proof generation cost may be fixed and per transaction cost.

Security and Finality:

Regarding security and finality, Optimistic Rollups needs time to resolve the dispute. Usually, it takes a few hours up to a week, which is why the times are longer. Security within the optimistic rollups is based on the incentivized validation of fraudulent transactions by the network validators within the dispute period. This makes this dispute mechanism entirely secure, but the longer finality time can be deemed disadvantageous when dealing with time-sensitive applications (Pope, 2016). On the other hand, ZK Rollups overcome this problem by providing an immediate finality with the cryptographic evidence of the transactions' finality. The high level of safety in Ethereum due to the employment of zero-knowledge proofs in ZK Rollups minimizes the possibility of malicious parties engaging in manipulations with the transactions. Since computations result in sound cryptographic verification, ZK Rollups is highly secure and suited for applications with improved user trust and rapid finality.

Additional Considerations:

Concerning complexity, Optimistic Rollups are more accessible to implement and also easier to interface with other blockchain systems because of the more straightforward challenge and response procedure. While compared with different kinds of Rollups, ZK Rollups are more complex to implement and update, and you need knowledge of cryptographic protocols and zero-knowledge proofs. Thus, in terms of the development and maintenance of Optimistic Rollups, there are apparent advantages to the simplicity of the development process and low overhead when it comes to maintenance. On the other hand, there is a higher initial investment and operational cost when it comes to ZK Rollups because of the complexity of coming up with and proving such proofs.

Use Cases:

This means that Optimistic Rollups can be used in almost any application due to their high throughput and low cost. Specifically, ZK Rollups are suitable for use cases that require better privacy and security and those that require transaction finality, such as private financial transactions and secure identity solutions.

FUTURE RESEARCH DIRECTIONS:

It is vital to understand that rollups are an entertaining sort of technology with some potential for further enhancement. Future research could focus on several areas:

Decentralization:

- **Increasing Validator Decentralization:** It is worth emphasizing that increasing decentralization for rollup validators can benefit safety and trust in the ecosystem. New ideas could be studied concerning encouraging more participants to act as validators and enhancing the stability of decentralized decision-making mechanisms.

- **Reducing Centralization Risks:** Search for ways to decrease the threats inherent to the centralization of the failure points, such as the fact that a single party can control a vast majority of the validators. This could include measures such as improving the openness and dictatorship of extra-validator selection procedures.

Interoperability:

- **Standardization:** Standardization is essential in achieving interoperability and interaction as a standard convention in rollup implementations and other Layer 2 solutions. This would allow for asset and data portability between different rollups and Layer 2 networks.
- **Cross-Rollup Communication Protocols:** Investigations into how such cross-rollup communication mechanisms would work could help improve the interactions between various rollup solutions and, thus, the scalability and functionality of the blockchain space.
- **Compatibility with Layer 1 Protocols:** Ensuring rollups can solve this problem would also mean less of a disconnect between Layer 1 blockchains and any other Layer 2 solutions that emerge in the future.

Usability:

- **User Experience (UX):** Bettering the rollup technology-friendly user interface is a perfect way to enhance the technology's uptake. This includes improving the user interfaces, shortening transaction times, and improving the information that end-users receive during the rollup process.
- **Developer Tools and Documentation:** Expanding the current support for Rollup developers' tools and documentation can reduce the existing barriers to application development in Rollups. Ironing out these issues could entail enhancing SDKs, providing clear documentation, and improving the general development platforms.
- **Onboarding and Education:** Activities such as raising users' and developers' awareness of the rollups and how they function may help increase their uptake. These could include developing literature on how to do business, conducting festivals, and offering practical lessons.

Efficiency:

- **Proof Generation and Verification:** Optimizing and verifying the proof layers in ZK Rollups to decrease costs is considered the most critical development step. This involves devising more efficient algorithms and additional computing methods, such as hardware acceleration, to hasten these processes.
- **Optimizing Data Storage:** Explore measures that will allow for efficient storage of data within rollups, such as better compression and off-chain solutions, in a bid to minimize the data size that is stored on-chain, hence cutting down on costs.
- **Reducing Computational Overheads:** Some of the future work involving rollup can be centred on reducing the time used in rollup computations, such as batch processing and generating proofs, which will ultimately be costly (Ozdemir et al., 2020).

Security:

- **Advanced Cryptographic Techniques:** Many options remain to be investigated; however, technologies like advanced cryptography can make rollups even more secure. This involves post-quantum cryptography to thwart future attacks from quantum computers.
- **Robust Fraud Detection:** Improving the Optimistic Rollups approach's fraud resilience without long challenge periods can make fraud more complex and increase security.

- **Privacy Enhancements:** We are developing solutions to enhance the privacy of ZK Rollups, which comprise better proofs for zero knowledge and highly avant-garde intelligent contracts that can enhance users' acceptance of these solutions for more private applications.

Scalability:

- **Layer 3 Solutions:** Exploring Layer 3 solutions that are based on top of rollups for even better scalability and enhanced lavish features. This could be rolled up hierarchically or by using other methods conducted off-chain.
- **Network Congestion Management:** Creating measures to avoid traffic jams in the working network and to successfully complete rollups during the rush(Al-Najada, 2016).

Governance:

- **Decentralized Governance Models:** Decentralization of rollup protocols so people can contribute to the structures and decision-making process that frame the rollup ecosystem's future.
- **Incentive Structures:** Investigating how to ensure that the right incentives are provided based on the participation rate of the validators and the network usage by the users (Restuccia et al., 2016).

Table 3 Comparison of Optimistic Rollups and ZK Rollups: Key Aspects

Aspect	Optimistic Rollups	ZK Rollups
Transaction Throughput	Thousands of TPS; scalability limited by on-chain data availability	Thousands of TPS; highly scalable due to batch processing and compact proofs; limited by proof creation cost
Transaction Costs	Lower transaction fees due to fewer on-chain computations; depends on transaction load and rollup type	Higher fees due to computational proof generation; still lower than Layer 1 transactions
Security and Finality	Security based on incentivized validation during dispute period; longer finality time (hours to a week)	Immediate finality with cryptographic proofs; high security due to zero-knowledge proofs
Complexity	Easier to implement and integrate; lower development and maintenance overhead	More complex to implement and update; requires advanced cryptographic knowledge; higher initial costs
Use Cases	Suitable for high throughput, low-cost applications; general-purpose use	Suitable for privacy-sensitive, security-focused, and rapid finality applications
Future Research Directions		
Decentralization	Increase validator decentralization, reduce centralization risks	Increase validator decentralization, reduce centralization risks
Interoperability	Standardization, cross-rollup communication protocols, compatibility with Layer 1 protocols	Standardization, cross-rollup communication protocols, compatibility with Layer 1 protocols
Usability	Improve user experience, developer tools, documentation, onboarding, and education	Improve user experience, developer tools, documentation, onboarding, and education
Efficiency	Optimize proof generation and verification, optimize data storage, reduce computational overheads	Optimize proof generation and verification, optimize data storage, reduce computational overheads
Security	Advanced cryptographic techniques, robust fraud detection, privacy enhancements	Advanced cryptographic techniques, robust fraud detection, privacy enhancements
Scalability	Explore Layer 3 solutions, network congestion management	Explore Layer 3 solutions, network congestion management
Governance	Develop decentralized governance models, investigate incentive structures	Develop decentralized governance models, investigate incentive structures

CONCLUSION

Rollups are seen as a breakthrough in scaling blockchains, as these issues are acute for Layer 1 blockchains, including Ethereum and Bitcoin. Optimistic Rollups and ZK Rollups are designed with strengths and weaknesses, so Optimistic is more suitable for some applications, while ZK is for others.

Optimistic rollups follow higher throughput and lower transaction fees, given that most transactions are legitimate and only challenged otherwise. Their easier implementation and relatively lower computation demand make them suitable for numerous use cases, especially if the finality time is longer, as is the case with several DeFi applications and game ecosystems.

ZK Rollups rely on complex computational formalities to provide zero-knowledge proofs to help achieve immutability and fast confirmation time with security. Also, due to their enhanced security, these numbers are best suited to use in instances that call for security and an immediate payout, such as obscure financial transactions and secure identification services. Due to the utilization of sophisticated zero-knowledge proofs, their deployment entails more significant costs in development and maintenance; nevertheless, this solution's advantages in terms of security and privacy are invaluable.

Rollups are set to be instrumental in the evolution of blockchain systems with more significant potential for growth, increased efficiency, and security shortly as R & D are being conducted. As for further development, the improvements in decentralization, interoperability, easy usage, better performance, and reliable security will affect the possibilities and areas of usage.



Figure 5 Optimistic vs. Zero-Knowledge Rollups

THE POTENTIAL OF ROLLUPS

1. **Scalability and Cost Efficiency:** Rollups enhance the interoperability of blockchain networks with high transaction capacity and affordable fees for thousands of transactions per second. This is necessary to ensure the popularization of blockchain technology and its application in fields such as finance, logistics, and the Internet of Things.
2. **Security Enhancements:** The security foundation is laid using zero-knowledge proofs in ZK Rollups. It translates to guaranteeing that the transactions occurring are valid and cannot be hacked. Optimistic Rollups, with built-in dispute mechanisms, also provide higher security, especially when they combine it with stimuli for validators.
3. **Broad Use Case Applicability:** Rollups are versatile and can be tweaked in many fields. From high-speed DeFi applications to private financial services, this new category aimed at meeting different demands will be equally beneficial for leveraging the core Ethereum network's capabilities.
4. **Improved User Experience:** As usability improvements are made, rollups will be easily accessible to developers and users. Improved user experience, more advanced development environments, and detailed documentation will assist in extending the rollup technology's usage further in existing and new applications.
5. **Sustainable Blockchain Growth:** Rollups help maintain a healthy growth model of blockchains by shifting transactions to sectors other than the central chain and improving data storage. They also assist in lowering problems based on traffic congestion and high fees that affect network users and developers.

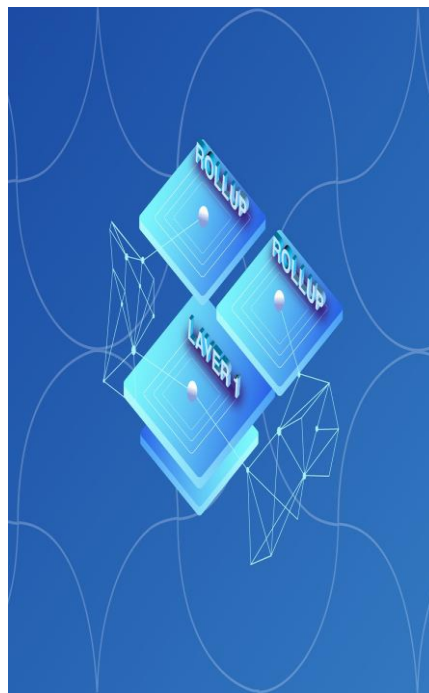


Figure 6 Advantages of Using Rollups Infrastructure for Blockchain Scalability

LOOKING FORWARD:

Future advancements in rollup technology research mean that blockchain networks can become more scalable, secure, and efficient. Further advancements in cryptography, mechanisms of governance, and the development of protocols in shared communication will continue to progress rollups as crucial players in the scaling up of blockchain technologies.

In the future, rollups will help the blockchain industry grow and, thus, allow the industry to overcome the present challenges. Rollups act as a tool to revolutionize blockchain technology use and integration with other systems, enhancing the utilization of blockchain technology and creating new decentralized applications.

Rollups are not just a solution to all blockchain scalability problems; they are a new age of improved blockchain networks. As this technology is further investigated and expanded, rollups will surely make an indispensable contribution to the creation of decentralized systems.

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